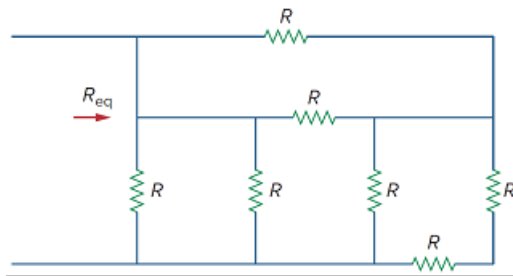


Problem

All resistors (R) in Fig. 2.93 are $10\ \Omega$ each. Find R_{eq} .

Figure 2.93:



Step-by-step solution

Step 1 of 3

Consider the circuit diagram of textbook Figure 2.93. The equivalent resistance of the bottom right of the circuit is,

$$\begin{aligned} [(R + R) \parallel R] &= \frac{(2R)(R)}{3R} \\ &= \frac{2}{3}R \end{aligned}$$

The resistors in the bottom left of the circuit are parallel and their equivalent resistance is,

$$\begin{aligned} (R \parallel R) &= \frac{(R)(R)}{2R} \\ &= \frac{1}{2}R \end{aligned}$$

The resultant circuit is as shown in Figure 1.

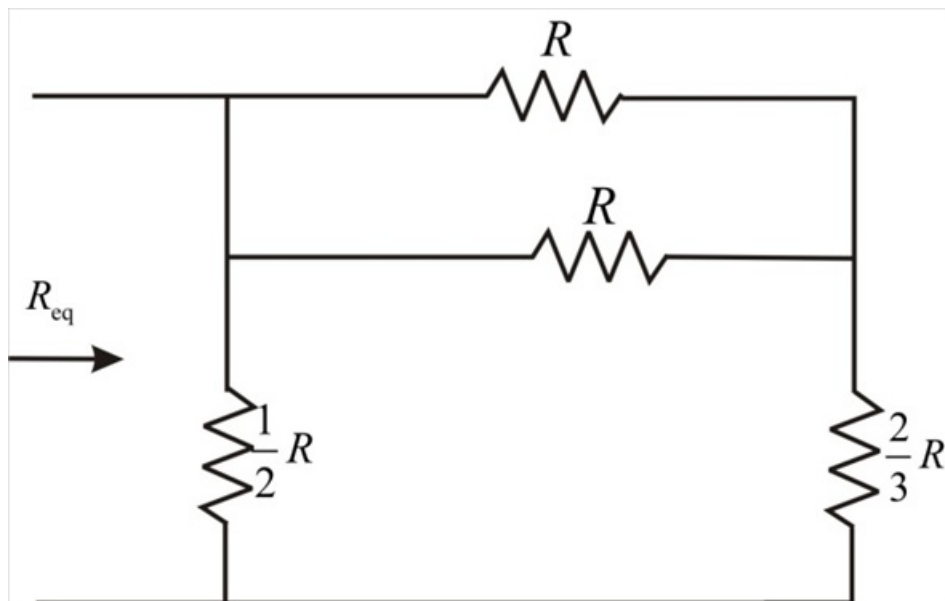


Figure 1

Step 2 of 3

From Figure 1, the upper two resistors are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned}(R \parallel R) &= \frac{(R)(R)}{2R} \\ &= \frac{1}{2}R\end{aligned}$$

The resultant circuit diagram is as shown in Figure 2.

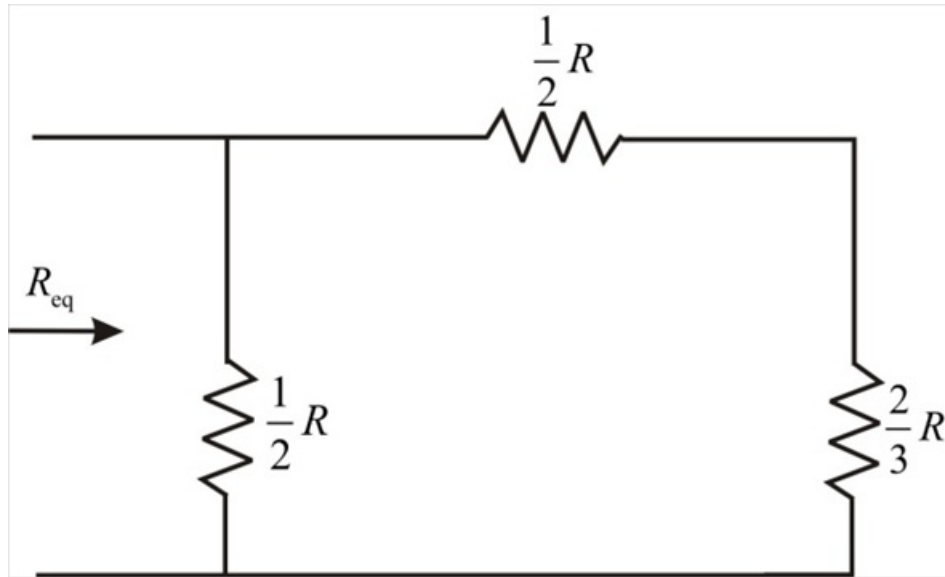


Figure 2

Step 3 of 3

From Figure 2, the equivalent resistance is,

$$\begin{aligned}R_{eq} &= \left(\frac{1}{2}R + \frac{2}{3}R \right) \parallel \frac{1}{2}R \\ &= \left(\frac{7}{6}R \right) \parallel \frac{1}{2}R \\ &= \frac{(7/12)R^2}{(10/6)R} \\ &= \frac{7R}{20}\end{aligned}$$

The resistance R is given as $10 \, \Omega$. Therefore the equivalent resistance is,

$$\begin{aligned}R_{eq} &= \frac{7(10)}{20} \\ &= 3.5 \, \Omega\end{aligned}$$

Therefore, the equivalent resistance of the textbook Figure 2.93 is $R_{eq} = 3.5 \, \Omega$.